

4.3. Thin Airfoil Theory

We have seen that an airfoil of arbitrary shape can be synthesized by combining a uniform flow with a curved vortex sheet wrapped around the surface of the airfoil. The strength of the sheet varies in such a way as to make the normal velocity component zero everywhere along the surface of the airfoil. In most cases, the strength distribution necessary to satisfy this condition is very difficult or impossible to determine analytically and numerical methods must be used. The vortex panel method described in the previous section is one such numerical method. Under certain conditions, however, it is possible to obtain an analytical solution. One such analytical solution is obtained by using an approximation known as “thin airfoil theory.”

For airfoils with a maximum thickness of about 12% or less, the inviscid aerodynamic forces and moments are only slightly affected by the thickness distribution. The resultant aerodynamic force and moment acting on such an airfoil depend almost exclusively on the angle of attack and shape of the camber line. For this reason, the inviscid aerodynamics for these airfoils can be closely approximated by assuming that the airfoil thickness is zero everywhere along the camber line. Thus, airfoils with a thickness of about 12% or less can be approximated by combining a uniform flow with a vortex sheet placed along the camber line, as is shown schematically in Fig. 4.3.1. The strength of this vortex sheet is allowed to vary with the distance, s , measured along the camber line. The variation in this strength, $\gamma(s)$, must be determined so that the camber line becomes a streamline for the flow and the Kutta condition is satisfied at the trailing edge.

Let the y -coordinate of the camber line, measured relative to the chord line, as a function of axial distance along the chord line from the leading edge, be denoted by $y_c(x)$. The angle between the chord line and the camber line is then given by

$$\phi = \tan^{-1}\left(\frac{dy_c}{dx}\right) \quad (4.3.1)$$

Let the tangential and normal components of the velocity induced by the vortex sheet be denoted as V_t and V_n , respectively. Then, from Fig.4.3.2, the x - and y -components of velocity resulting from both the uniform flow and the vortex sheet are

$$V_x = V_\infty \cos \alpha + V_t \cos \phi - V_n \sin \phi \quad (4.3.2)$$

$$V_y = V_\infty \sin \alpha + V_t \sin \phi + V_n \cos \phi \quad (4.3.3)$$

If the camber line is to be a streamline for the flow, we must have

$$\frac{dy_c}{dx} = \frac{V_y}{V_x} \quad (4.3.4)$$

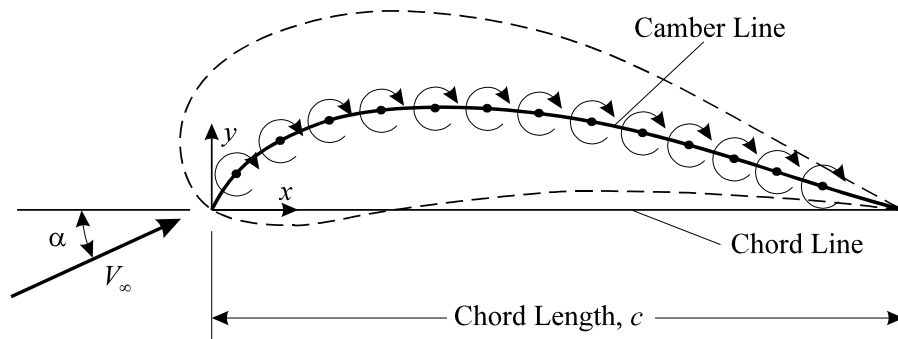


Figure 4.3.1. Synthesis of a thin airfoil section from a uniform flow and a curved vortex sheet distributed along the camber line.

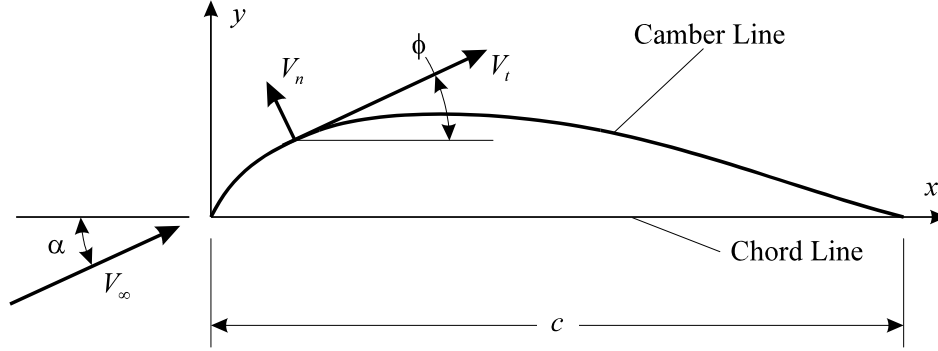


Figure 4.3.2. Determination of the x - and y -components of velocity induced by the vortex sheet and the uniform flow.

or after applying Eqs. (4.3.2) and (4.3.3) and rearranging,

$$(V_{\infty} \cos \alpha + V_t \cos \phi - V_n \sin \phi) \frac{dy_c}{dx} = V_{\infty} \sin \alpha + V_t \sin \phi + V_n \cos \phi \quad (4.3.5)$$

The approximation used in thin airfoil theory is that both α and ϕ are small enough so that we can neglect terms of order α^2 and ϕ^2 when compared with terms of order α and ϕ . Using this approximation in the Taylor series expansions for sine and cosine we have the classical small angle approximation

$$\cos \alpha \cong \cos \phi \cong 1 \quad (4.3.6)$$

$$\sin \alpha \cong \alpha \quad (4.3.7)$$

$$\sin \phi \cong \phi \cong \tan \phi = \frac{dy_c}{dx} \quad (4.3.8)$$

Using these approximations in Eq. (4.3.5), we have

$$\left(V_{\infty} + V_t - V_n \frac{dy_c}{dx} \right) \frac{dy_c}{dx} = V_{\infty} \alpha + V_t \frac{dy_c}{dx} + V_n \quad (4.3.9)$$

Solving Eq. (4.3.9) for V_n and neglecting the second order term gives

$$V_n = V_{\infty} \left(\frac{dy_c}{dx} - \alpha \right) \quad (4.3.10)$$

Typically, airfoil sections have a maximum camber on the order of about 2% of the chord length. Thus, the camber line is always very close to the chord line, and the normal velocity induced by a vortex sheet placed on the camber line is very nearly the same as that for a vortex sheet placed on the chord line. Thus, using Eq. (3.12.6) with $y = 0$ and changing the limits of integration to correspond to a sheet extending from $x_o = 0$ to $x_o = c$, we have

$$V_n = -\frac{1}{2\pi} \int_{x_o=0}^c \frac{\gamma(x_o)}{x - x_o} dx_o \quad (4.3.11)$$

Using Eq. (4.3.11) in Eq. (4.3.10) we have

$$\frac{1}{2\pi} \int_{x_o=0}^c \frac{\gamma(x_o)}{x-x_o} dx_o = V_\infty \left(\alpha - \frac{dy_c}{dx} \right) \quad (4.3.12)$$

Equation (4.3.12) is the fundamental equation of thin airfoil theory. Any vortex strength distribution, $\gamma(x_o)$, which satisfies Eq. (4.3.12) will make the camber line a streamline of the flow, at least within the accuracy of the approximations that we have used. For a given airfoil at a given angle of attack, the only unknown in Eq. (4.3.12) is the vortex strength distribution, $\gamma(x_o)$. Thin airfoil solutions involve solving Eq. (4.3.12) for $\gamma(x_o)$ subject to the Kutta condition, which requires

$$\gamma(c) = 0 \quad (4.3.13)$$

The simplest solution to Eq. (4.3.12) is for the case of a **symmetric airfoil**. For such an airfoil, the camber line is coincident with the chord line and the change in the y -coordinate of the camber line with respect to the axial coordinate is zero at every point along the chord. Thus, from Eq. (4.3.12), for the symmetric airfoil we require

$$\frac{1}{2\pi} \int_{x_o=0}^c \frac{\gamma(x_o)}{x-x_o} dx_o = V_\infty \alpha \quad (4.3.14)$$

Equation (4.3.14) can be solved using the change of variables

$$x \equiv \frac{c}{2}(1 - \cos \theta) \quad (4.3.15)$$

which gives

$$x_o \equiv \frac{c}{2}(1 - \cos \theta_o) \quad (4.3.16)$$

and

$$dx_o = \frac{c}{2} \sin \theta_o d\theta_o \quad (4.3.17)$$

Solving Eq. (4.3.15) for θ , we can write θ as a function of x ,

$$\theta(x) = \cos^{-1} \left(1 - 2\frac{x}{c} \right) \quad (4.3.18)$$

So we have

$$\theta(0) = 0 \quad (4.3.19)$$

and

$$\theta(c) = \pi \quad (4.3.20)$$

Using this change of variables in Eq. (4.3.14) gives

$$\frac{1}{2\pi} \int_{\theta_o=0}^{\pi} \frac{\gamma(\theta_o) \sin \theta_o}{\cos \theta_o - \cos \theta} d\theta_o = V_\infty \alpha \quad (4.3.21)$$

The key to solving Eq. (4.3.21), and the reason for making the change of variables given by Eq. (4.3.15), is the existence of the integral relation

$$\int_{\theta_o=0}^{\pi} \frac{\cos(n\theta_o)}{\cos\theta_o - \cos\theta} d\theta_o = \frac{\pi \sin(n\theta)}{\sin\theta}, \quad n = 0, 1, 2, 3, 4, \dots \quad (4.3.22)$$

Notice that the result of this integral is independent of θ for both $n = 0$ and $n = 1$. Thus, we shall let

$$\frac{\gamma(\theta_o) \sin\theta_o}{\cos\theta_o - \cos\theta} = C_0 \frac{1}{\cos\theta_o - \cos\theta} + C_1 \frac{\cos\theta_o}{\cos\theta_o - \cos\theta} \quad (4.3.23)$$

Using Eq. (4.3.23) in Eq. (4.3.21) gives

$$\frac{C_0}{2\pi} \int_{\theta_o=0}^{\pi} \frac{1}{\cos\theta_o - \cos\theta} d\theta_o + \frac{C_1}{2\pi} \int_{\theta_o=0}^{\pi} \frac{\cos\theta_o}{\cos\theta_o - \cos\theta} d\theta_o = V_{\infty} \alpha \quad (4.3.24)$$

After applying Eq. (4.3.22), Eq. (4.3.24) reduces to

$$\frac{C_0}{2\pi}(0) + \frac{C_1}{2\pi}(\pi) = V_{\infty} \alpha \quad (4.3.25)$$

which results in

$$C_1 = 2V_{\infty} \alpha \quad (4.3.26)$$

Using Eq. (4.3.26) in Eq. (4.3.23), the solution to Eq. (4.3.21) is given by

$$\frac{\gamma(\theta_o) \sin\theta_o}{\cos\theta_o - \cos\theta} = \frac{C_0 + 2V_{\infty} \alpha \cos\theta_o}{\cos\theta_o - \cos\theta} \quad (4.3.27)$$

or

$$\gamma(\theta_o) = \frac{C_0 + 2V_{\infty} \alpha \cos\theta_o}{\sin\theta_o} \quad (4.3.28)$$

Applying the Kutta condition from Eq. (4.3.13), along with (4.3.20), we have

$$\gamma(c) = \frac{C_0 + 2V_{\infty} \alpha \cos\pi}{\sin\pi} = 0 \quad (4.3.29)$$

which requires

$$C_0 = -2V_{\infty} \alpha \quad (4.3.30)$$

Using Eq. (4.3.30) in Eq. (4.3.28), the vortex strength distribution that can be used to synthesize a thin symmetric airfoil section is given by

$$\gamma(\theta_o) = 2V_{\infty} \alpha \frac{1 + \cos\theta_o}{\sin\theta_o} \quad (4.3.31)$$

or using Eq. (4.3.16)

$$\gamma(x_o) = 2V_\infty \alpha \sqrt{\frac{c}{x_o} - 1} \quad (4.3.32)$$

Now, for the more general case of a **cambered airfoil**, we shall also try using the change of variables defined by Eq. (4.3.15). Using this change of variables in Eq. (4.3.12), the general equation for thin cambered airfoils is

$$\frac{1}{2\pi} \int_{\theta_o=0}^{\pi} \frac{\gamma(\theta_o) \sin \theta_o}{\cos \theta_o - \cos \theta} d\theta_o = V_\infty \left(\alpha - \frac{dy_c}{dx} \right) \quad (4.3.33)$$

A solution to this equation is found by adding a Fourier series to a term of the same form as the solution found for the thin symmetric airfoil. Thus, we assume a solution for the vortex strength distribution in the form

$$\gamma(\theta_o) = 2V_\infty \left(A_0 \frac{1 + \cos \theta_o}{\sin \theta_o} + \sum_{n=1}^{\infty} A_n \sin(n\theta_o) \right) \quad (4.3.34)$$

Equation (4.3.34) is not a solution to Eq. (4.3.33) unless we are able to determine a set of coefficients, A_n , such that Eq. (4.3.33) is satisfied. At this point, Eq. (4.3.34) simply represents a definition for the coefficient set A_n . Since each term in Eq. (4.3.34) is zero when $\theta_o = \pi$, the Kutta condition is directly satisfied by this definition.

Using Eq. (4.3.34) in Eq. (4.3.33) we have

$$\frac{A_0}{\pi} \int_{\theta_o=0}^{\pi} \frac{1 + \cos \theta_o}{\cos \theta_o - \cos \theta} d\theta_o + \sum_{n=1}^{\infty} \frac{A_n}{\pi} \int_{\theta_o=0}^{\pi} \frac{\sin(n\theta_o) \sin \theta_o}{\cos \theta_o - \cos \theta} d\theta_o = \alpha - \frac{dy_c}{dx} \quad (4.3.35)$$

The integrals in Eq. (4.3.35) can be evaluated to give

$$\int_{\theta_o=0}^{\pi} \frac{1 + \cos \theta_o}{\cos \theta_o - \cos \theta} d\theta_o = \pi \quad (4.3.36)$$

$$\int_{\theta_o=0}^{\pi} \frac{\sin(n\theta_o) \sin \theta_o}{\cos \theta_o - \cos \theta} d\theta_o = -\pi \cos(n\theta) \quad (4.3.37)$$

Using Eqs. (4.3.36) and (4.3.37) in Eq. (4.3.35) gives

$$A_0 - \sum_{n=1}^{\infty} A_n \cos(n\theta) = \alpha - \frac{dy_c}{dx} \quad (4.3.38)$$

The left-hand side of Eq. (4.3.38) is a standard Fourier series expansion for the known function on the right. Thus, the coefficients are given by

$$A_0 = \frac{1}{\pi} \int_{\theta=0}^{\pi} \left(\alpha - \frac{dy_c}{dx} \right) d\theta = \alpha - \frac{1}{\pi} \int_{\theta=0}^{\pi} \frac{dy_c}{dx} d\theta \quad (4.3.39)$$

$$A_n = -\frac{2}{\pi} \int_{\theta=0}^{\pi} \left(\alpha - \frac{dy_c}{dx} \right) \cos(n\theta) d\theta = \frac{2}{\pi} \int_{\theta=0}^{\pi} \frac{dy_c}{dx} \cos(n\theta) d\theta \quad (4.3.40)$$

Notice that the first coefficient, A_0 , is a function of both the angle of attack and the shape of the camber line. However, the remaining coefficients are functions only of the shape of the camber line. Using Eqs. (4.3.39) and (4.3.40) in Eq. (4.3.34) provides a solution to Eq. (4.3.33).

The Kutta-Joukowski law can be used to compute the lift per unit span for the entire airfoil section or for any axial segment of the airfoil. Thus we can write

$$\tilde{L} = \rho V_\infty \Gamma = \rho V_\infty \int_{x_o=0}^c \gamma(x_o) dx_o = \frac{\rho V_\infty c}{2} \int_{\theta_o=0}^{\pi} \gamma(\theta_o) \sin \theta_o d\theta_o \quad (4.3.41)$$

Using Eq. (4.3.34) in Eq. (4.3.41), we have

$$\tilde{L} = \rho V_\infty^2 c \left(A_0 \int_{\theta_o=0}^{\pi} (1 + \cos \theta_o) d\theta_o + \sum_{n=1}^{\infty} A_n \int_{\theta_o=0}^{\pi} \sin(n\theta_o) \sin \theta_o d\theta_o \right) \quad (4.3.42)$$

The integrals in Eq. (4.3.42) can be evaluated to give

$$\int_{\theta_o=0}^{\pi} (1 + \cos \theta_o) d\theta_o = \pi \quad (4.3.43)$$

and

$$\int_{\theta_o=0}^{\pi} \sin(n\theta_o) \sin \theta_o d\theta_o = \begin{cases} \pi/2, & \text{for } n=1 \\ 0, & \text{for } n \neq 1 \end{cases} \quad (4.3.44)$$

Thus, Eq. (4.3.42) becomes

$$\tilde{L} = \rho V_\infty^2 c \left(\pi A_0 + \frac{\pi}{2} A_1 \right) \quad (4.3.45)$$

The section lift coefficient is then

$$\tilde{C}_L \equiv \frac{\tilde{L}}{\frac{1}{2} \rho V_\infty^2 c} = 2\pi \left(A_0 + \frac{1}{2} A_1 \right) \quad (4.3.46)$$

Using Eqs. (4.3.39) and (4.3.40), Eq. (4.3.46) becomes

$$\tilde{C}_L = 2\pi \left(\alpha - \frac{1}{\pi} \int_{\theta=0}^{\pi} \frac{dy_c}{dx} (1 - \cos \theta) d\theta \right) = 2\pi(\alpha - \alpha_{L0}) \quad (4.3.47)$$

where α_{L0} is the zero lift angle of attack. Note, from Eq. (4.3.47), that thin airfoil theory predicts a section lift coefficient that is a linear function of angle of attack, and that the change in lift coefficient with respect to angle of attack is 2π per radian. Also note that the lift coefficient at zero angle of attack is a function only of the shape of the camber line. Thus, for thin airfoils,

$$\text{section lift slope} \equiv \frac{d\tilde{C}_L}{d\alpha} \equiv \tilde{C}_{L,\alpha} = 2\pi \quad (4.3.48)$$

and

$$\text{zero lift angle of attack} \equiv \alpha_{L0} = \frac{1}{\pi} \int_{\theta=0}^{\pi} \frac{dy_c}{dx} (1 - \cos \theta) d\theta \quad (4.3.49)$$

From the Kutta-Joukowski law, the section lift acting on a differential segment of the airfoil located at $x=x_o$ and having a length dx_o is

$$d\tilde{L} = \rho V_\infty d\Gamma = \rho V_\infty \gamma(x_o) dx_o \quad (4.3.50)$$

Within the approximations used for thin airfoils, the moment arm for this differential segment relative to the leading edge is x_o . Thus, the section moment about the leading edge is

$$\tilde{m}_{le} = -\rho V_\infty \int_{x_o=0}^c \gamma(x_o) x_o dx_o = -\frac{\rho V_\infty c^2}{4} \int_{\theta_o=0}^{\pi} \gamma(\theta_o) (1 - \cos \theta_o) \sin \theta_o d\theta_o \quad (4.3.51)$$

Using Eq. (4.3.34) in Eq. (4.3.51) gives

$$\tilde{m}_{le} = -\frac{\rho V_\infty^2 c^2}{2} \left(A_0 \int_{\theta_o=0}^{\pi} (1 - \cos^2 \theta_o) d\theta_o + \sum_{n=1}^{\infty} A_n \int_{\theta_o=0}^{\pi} (1 - \cos \theta_o) \sin(n\theta_o) \sin \theta_o d\theta_o \right) \quad (4.3.52)$$

The integrals in Eq. (4.3.52) can be evaluated to give

$$\int_{\theta_o=0}^{\pi} (1 - \cos^2 \theta_o) d\theta_o = \frac{\pi}{2} \quad (4.3.53)$$

$$\int_{\theta_o=0}^{\pi} \sin(n\theta_o) \sin \theta_o d\theta_o = \begin{cases} \pi/2, & \text{for } n=1 \\ 0, & \text{for } n \neq 1 \end{cases} \quad (4.3.54)$$

$$\int_{\theta_o=0}^{\pi} \cos \theta_o \sin(n\theta_o) \sin \theta_o d\theta_o = \frac{1}{2} \int_{\theta_o=0}^{\pi} \sin(n\theta_o) \sin(2\theta_o) d\theta_o = \begin{cases} \pi/4, & \text{for } n=2 \\ 0, & \text{for } n \neq 2 \end{cases} \quad (4.3.55)$$

Using Eqs. (4.3.53) through (4.3.55) in Eq. (4.3.52), we obtain

$$\tilde{m}_{le} = -\frac{\pi \rho V_\infty^2 c^2}{4} \left(A_0 + A_1 - \frac{1}{2} A_2 \right) \quad (4.3.56)$$

Thus, the section moment coefficient about the leading edge is then

$$\tilde{C}_{m_{le}} \equiv \frac{\tilde{m}_{le}}{\frac{1}{2} \rho V_\infty^2 c^2} = -\frac{\pi}{2} \left(A_0 + A_1 - \frac{1}{2} A_2 \right) \quad (4.3.57)$$

From Eq. (4.3.46) we can write

$$A_0 = \frac{\tilde{C}_L}{2\pi} - \frac{A_1}{2} \quad (4.3.58)$$

Using Eq. (4.3.58) in Eq. (4.3.57) we have

$$\tilde{C}_{m_{le}} = -\frac{\tilde{C}_L}{4} + \frac{\pi}{4} (A_2 - A_1) \quad (4.3.59)$$

Using Eq. (4.3.40) in Eq. (4.3.59) results in

$$\tilde{C}_{m_{le}} = -\frac{\tilde{C}_L}{4} + \frac{1}{2} \int_{\theta=0}^{\pi} \frac{dy_c}{dx} [\cos(2\theta) - \cos\theta] d\theta \quad (4.3.60)$$

The section moment coefficient about the quarter-chord is then

$$\tilde{C}_{m_{c/4}} = \tilde{C}_{m_{le}} + \frac{\tilde{C}_L}{4} = \frac{\pi}{4} (A_2 - A_1) \quad (4.3.61).$$

or using Eq. (4.3.40)

$$\tilde{C}_{m_{c/4}} = \frac{1}{2} \int_{\theta=0}^{\pi} \frac{dy_c}{dx} [\cos(2\theta) - \cos\theta] d\theta \quad (4.3.62)$$

The center of pressure can be computed from

$$\frac{x_{cp}}{c} = -\frac{\tilde{m}_{le}}{c \tilde{L}} = -\frac{\tilde{C}_{m_{le}}}{\tilde{C}_L} \quad (4.3.63)$$

Using Eq. (4.3.59) in Eq. (4.3.63) results in

$$\frac{x_{cp}}{c} = \frac{1}{4} + \frac{\pi}{4\tilde{C}_L} (A_1 - A_2) \quad (4.3.64)$$

or in view of Eq. (4.3.40)

$$\frac{x_{cp}}{c} = \frac{1}{4} + \frac{1}{2\tilde{C}_L} \int_{\theta=0}^{\pi} \frac{dy_c}{dx} [\cos\theta - \cos(2\theta)] d\theta \quad (4.3.65)$$

Note that the leading edge moment coefficient and the center of pressure both depend on the lift coefficient and hence on the angle of attack. The moment about the leading edge becomes more positive and the center of pressure moves farther back as the angle of attack and the lift coefficient are decreased. In fact as the angle of attack approaches the zero lift angle of attack, the center of pressure moves off the airfoil and goes to infinity. The quarter-chord moment coefficient, on the other hand, is independent of the angle of attack and depends only on the shape of the camber line. Since the aerodynamic center is defined to be the point on the airfoil where the moment is independent of angle of attack, we see that the quarter-chord is the aerodynamic center for thin airfoils. Thus, the quarter-chord is usually referred to as the theoretical or ideal aerodynamic center of the cambered airfoil. Viscous effects and airfoil thickness can cause the quarter-chord moment coefficient to vary slightly with angle of attack, but this variation is small and the aerodynamic center is always near the quarter-chord. For this reason, airfoil section moment data are usually reported in terms of the quarter-chord moment coefficient.

For the special case of a symmetric airfoil section, from Eqs. (4.3.47), (4.3.60), (4.3.62) and (4.3.65), the section lift coefficient, the leading edge moment coefficient, the quarter-chord moment coefficient, and the center of pressure for a **symmetric airfoil section** are respectively,

$$\tilde{C}_L = 2\pi\alpha \quad (4.3.66)$$

$$\tilde{C}_{m_{le}} = -\frac{\tilde{C}_L}{4} \quad (4.3.67)$$

$$\tilde{C}_{m_{c/4}} = 0 \quad (4.3.68)$$

$$\frac{x_{cp}}{c} = \frac{1}{4} \quad (4.3.69)$$

Thus we see that, for a thin symmetric airfoil section, the zero lift angle of attack is exactly zero. So the section lift coefficient for a thin symmetric airfoil is simply 2π multiplied by the angle of attack in radians. We also see that the quarter-chord is not only the aerodynamic center of a thin symmetric airfoil section, but it is also the center of pressure.

As is the case with all two-dimensional potential flow, thin airfoil theory predicts a net aerodynamic force that is normal to the freestream. Thus, thin airfoil theory predicts a section drag coefficient that is exactly zero. This is not a function of the thin airfoil approximation. The numerical panel method discussed in the previous section will also predict zero section drag, including the effects of thickness. Section drag in a two-dimensional flow results entirely from viscous effects, which are neglected in the potential flow equations. The viscous forces also have some effect on the lift, but this effect is relatively small.

Summary

A thin airfoil can be synthesized from a uniform flow combined with a vortex sheet placed on the camber line. The **vortex strength** varies according to the relation,

$$\gamma(\theta) = 2V_\infty \left(A_0 \frac{1 + \cos \theta}{\sin \theta} + \sum_{n=1}^{\infty} A_n \sin(n\theta) \right)$$

where

$$A_0 = \alpha - \frac{1}{\pi} \int_{\theta=0}^{\pi} \frac{dy_c}{dx} d\theta$$

$$A_n = \frac{2}{\pi} \int_{\theta=0}^{\pi} \frac{dy_c}{dx} \cos(n\theta) d\theta$$

$$x(\theta) \equiv \frac{c}{2} (1 - \cos \theta)$$

The **section lift and moment coefficients and the center of pressure** for a thin airfoil are given by,

$$\tilde{C}_L = 2\pi(A_0 + \frac{1}{2}A_1) = 2\pi \left(\alpha - \frac{1}{\pi} \int_{\theta=0}^{\pi} \frac{dy_c}{dx} (1 - \cos \theta) d\theta \right) = 2\pi(\alpha - \alpha_{L0})$$

$$\tilde{C}_{m_{le}} = -\frac{\tilde{C}_L}{4} + \frac{\pi}{4}(A_2 - A_1) = -\frac{\tilde{C}_L}{4} + \frac{1}{2} \int_{\theta=0}^{\pi} \frac{dy_c}{dx} [\cos(2\theta) - \cos \theta] d\theta$$

$$\tilde{C}_{m_{c/4}} = \frac{\pi}{4}(A_2 - A_1) = \frac{1}{2} \int_{\theta=0}^{\pi} \frac{dy_c}{dx} [\cos(2\theta) - \cos \theta] d\theta$$

$$\frac{x_{cp}}{c} = \frac{1}{4} + \frac{\pi}{4\tilde{C}_L}(A_1 - A_2) = \frac{1}{4} + \frac{1}{2\tilde{C}_L} \int_{\theta=0}^{\pi} \frac{dy_c}{dx} [\cos \theta - \cos(2\theta)] d\theta$$